

$$(a) \quad x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega) e^{j\omega t} d\omega \quad \text{معادله 1}$$

$$F \quad : \quad X(j\omega) = \delta(\omega - \omega_0) \quad \text{برای}$$

$$F^{-1} \{ \delta(\omega - \omega_0) \} = \frac{1}{2\pi} e^{j\omega_0 t}$$

$$\Rightarrow F^{-1} \{ 3\delta(\omega - 1) + j2\delta(\omega - 2) + 3\delta(\omega + 1) - j2\delta(\omega + 2) \}$$

$$= \frac{1}{2\pi} \left[3e^{jt} + j2e^{j2t} + 3e^{-jt} - j2e^{-j2t} \right]$$

$$e^{jt} + e^{-jt} = 2\cos t \quad \text{از طرفین ضرب در 3}$$

$$e^{j2t} - e^{-j2t} = 2j\sin 2t$$

$$\Rightarrow F^{-1} \{ X(j\omega) \} = \frac{1}{2\pi} [6\cos t - 4\sin 2t]$$

$$\Rightarrow \boxed{x(t) = \frac{3}{\pi} \cos t - \frac{2}{\pi} \sin 2t}$$

(b)

$$\sin \frac{\omega}{2} = \frac{1}{2j} \left(e^{j\frac{\omega}{2}} - e^{-j\frac{\omega}{2}} \right) \quad \text{مساوی}$$

برای $\rightarrow$ $\frac{\sin(\frac{\omega}{2})}{j\omega + 2} e^{-j2\omega} = \frac{1}{2j} \left[\frac{e^{j\omega/2}}{j\omega + 2} - \frac{e^{-j\omega/2}}{j\omega + 2} \right] e^{-j2\omega}$

$$= \frac{1}{2j} \left[\frac{e^{-j3/2\omega}}{j\omega + 2} - \frac{e^{-j5/2\omega}}{j\omega + 2} \right]$$

$$\frac{1}{j\omega + a} \xleftrightarrow{F} e^{-at} u(t) \quad \text{مساوی}$$

$$e^{-j\omega t_0} H(j\omega) \iff h(t - t_0)$$

برای $\rightarrow$ $e^{-j\omega 3/2} \frac{1}{j\omega + 2} \iff e^{-2(t-3/2)} u(t-3/2)$

$$\frac{e^{-j\omega 5/2}}{j\omega + 2} \iff e^{-2(t-5/2)} u(t-5/2)$$

~~برای $\rightarrow$ $\frac{1}{j\omega + 2} \iff e^{-2t} u(t)$~~

$$\Rightarrow X(j\omega) \iff \frac{1}{2j} \left[e^{-2(t-3/2)} u(t-3/2) - e^{-2(t-5/2)} u(t-5/2) \right]$$

$$(c) \quad X(j\omega) = \frac{3(3+j\omega)}{(4+j\omega)(2+j\omega)} = \frac{A}{j\omega+2} + \frac{B}{j\omega+4}$$

$$\Rightarrow A = \frac{3}{2}, \quad B = \frac{3}{2}$$

$$\text{by } \mathcal{L}^{-1} \rightarrow X(j\omega) = \frac{3}{2} \left[\frac{1}{j\omega+2} + \frac{1}{j\omega+4} \right]$$

$$\frac{1}{j\omega+a} \Leftrightarrow e^{-at} u(t) \quad \text{by } \mathcal{L}$$

$$\Rightarrow x(t) = \frac{3}{2} [e^{-2t} + e^{-4t}] u(t)$$

$$(d) \quad X(j\omega) = \frac{-(j\omega)^2 + 4(j\omega) + 6}{[(j\omega)^2 + 3j\omega + 2](j\omega+4)}$$

$$= \frac{-[(j\omega)^2 + 4j\omega + 6]}{(j\omega+1)(j\omega+2)(j\omega+4)}$$

$$= \frac{A}{j\omega+1} + \frac{B}{j\omega+2} + \frac{C}{j\omega+4}$$

$$\Rightarrow A = C = -1, \quad B = 1$$

از طرفی می دانیم $\Rightarrow \frac{1}{j\omega + a} \Leftrightarrow e^{-at} u(t), a > 0$

$$\Rightarrow x(t) = (-e^{-t} + e^{-2t} - e^{-4t}) u(t)$$

(e) $x(j\omega) = \frac{2j\omega + 1}{(j\omega + 2)^2} = \frac{A}{j\omega + 2} + \frac{B}{(j\omega + 2)^2}$

~~1~~ $\frac{x}{(j\omega + 2)^2} \Rightarrow 2j\omega + 1 = A(j\omega + 2) + B$

$$\Rightarrow A = 2, B = -3$$

$$\Rightarrow x(j\omega) = \frac{2}{j\omega + 2} - \frac{3}{(j\omega + 2)^2}$$

می دانیم: $\frac{1}{(j\omega + a)^2} \xLeftrightarrow F t e^{-at} u(t)$

$$\Rightarrow x(t) = 2e^{-2t} u(t) - 3te^{-2t} u(t)$$

$$\Rightarrow x(t) = (2 - 3t)e^{-2t} u(t)$$

$$(P) \quad X(j\omega) = \frac{4 \sin^2(\omega)}{\omega^2} = \left(\frac{2 \sin \omega}{\omega} \right)^2$$

$$\Rightarrow \frac{2 \sin \omega}{\omega} \iff x_p(t) = \begin{cases} 1 & -1 < t < 1 \\ 0 & \text{o. w.} \end{cases}$$

ضرب در محدوده فرکانسها را کونولوشن در حوزه

زمان است.

$$x(t) = x_p(t) * x_p(t)$$

conv که پالس عرض 2 در خودش یک مثلث با قاعده 4

$$x(t) = \begin{cases} 2 - |t| & , |t| \leq 2 \\ 0 & , \text{o. w.} \end{cases}$$

$$\Rightarrow x(t) = (2 - |t|) [u(t+2) - u(t-2)]$$

$$(g) X(j\omega) = \frac{d}{d\omega} \underbrace{\left[4 \cos 3\omega \frac{\sin 2\omega}{\omega} \right]}_{Y(j\omega)}$$

$$\frac{2 \sin(T\omega)}{\omega} \xLeftrightarrow \text{از طریق فیلتر} \text{rectangle } (|t| < T)$$

$$\Rightarrow \frac{\sin 2\omega}{\omega} \Leftrightarrow \frac{1}{2} \text{rectangle } (|t| < 2)$$

فرض: $a(t) = \frac{1}{2} \text{rectangle } (|t| < 2)$

از طریق: $\cos 3\omega = \frac{1}{2} (e^{j3\omega} + e^{-j3\omega})$

~~از طریق فیلتر~~

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$$\cos(3\omega) \frac{\sin 2\omega}{\omega} = \frac{1}{2} \left[e^{j3\omega} \frac{\sin 2\omega}{\omega} + e^{-j3\omega} \frac{\sin 2\omega}{\omega} \right]$$

$$\xLeftrightarrow^F \frac{1}{2} [a(t-3) + a(t+3)]$$

$$Y(j\omega) = 4 \cos 3\omega \frac{\sin 2\omega}{\omega} \quad \text{حال}$$

$$\xleftrightarrow{F} y(t) = 2 [a(t-3) + a(t+3)]$$

$$\Rightarrow y(t) = \begin{cases} 1 & , 1 < t < 5 \\ 1 & , -5 < t < -1 \\ 0 & \text{c.w.} \end{cases}$$

حال با توجه به قانون مشتق :

$$t y(t) \xleftrightarrow{F} j \frac{d}{d\omega} Y(j\omega)$$

$$j \frac{d}{d\omega} Y(j\omega) \xleftrightarrow{F^{-1}} -j t y(t)$$

$$\hat{\Rightarrow} x(t) = \begin{cases} -jt & -5 < t < -1 \\ -jt & 1 < t < 5 \\ 0 & \text{c.w.} \end{cases}$$

$$\Rightarrow x(t) = -jt [u(t-1) - u(t-5) + u(t+5) - u(t+1)]$$

سوال 2/5 :

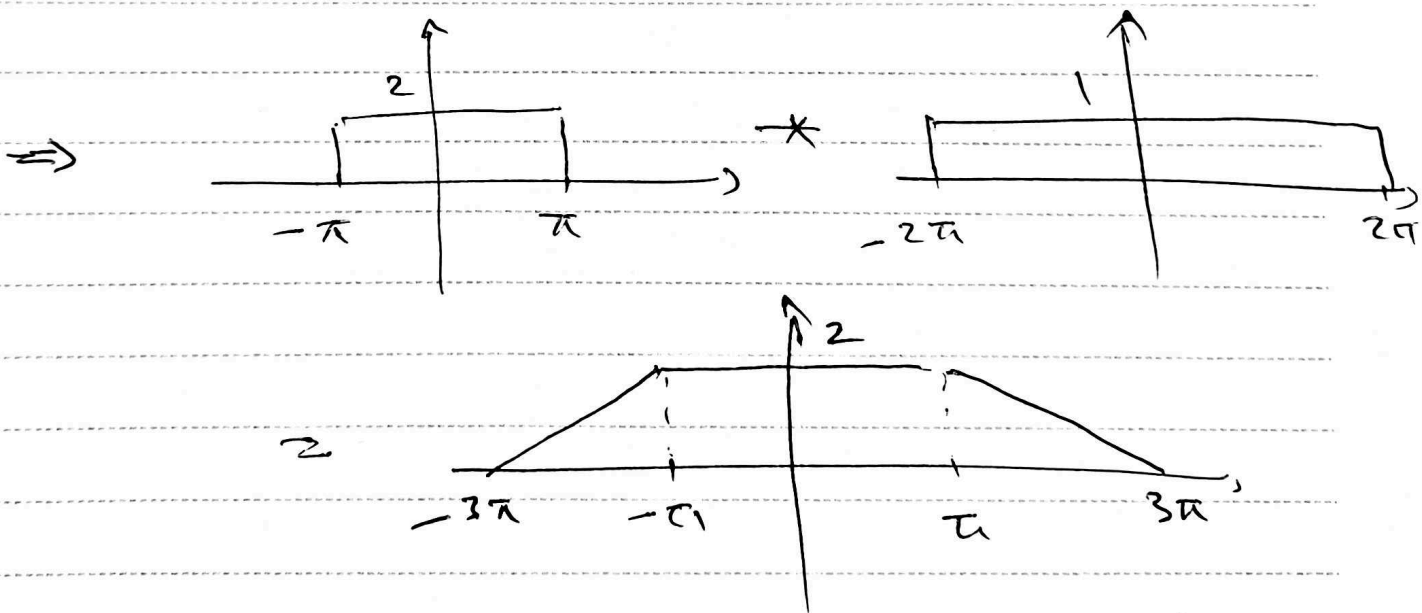
$$(a) \quad x(t) = \underbrace{\left[\frac{2 \sin(\pi t)}{\pi t} \right]}_{x_1(t)} \underbrace{\left[\frac{\sin(2\pi t)}{\pi t} \right]}_{x_2(t)}$$

$$x_1(t) \iff X_1(j\omega) = 2, \quad |\omega| < \pi$$

$$x_2(t) \iff X_2(j\omega) = 1, \quad |\omega| < 2\pi$$

هر تابع ضرب در عرضی ~~و~~ زمان ~~و~~ کانولوشن در عرضی
فرکانس است:

$$X(j\omega) = \frac{1}{2\pi} X_1 * X_2$$



$$X(j\omega) = \begin{cases} 2 & |\omega| \leq \pi \\ \frac{3\pi - |\omega|}{\pi} & \pi < |\omega| < 3\pi \\ 0 & |\omega| \geq 3\pi \end{cases}$$

$$(b) \quad x(t) = \frac{1}{1+jt}$$

با استفاده از مکزن دومانز: $e^{-t} u(t) \Leftrightarrow \frac{1}{1+j\omega}$

$$x(t) \Leftrightarrow X(j\omega)$$

$$X(t) \Leftrightarrow 2\pi x(-\omega)$$

حالا بر این سوال: $x(t) = \frac{1}{1+jt}$

$$X(j\omega) = \begin{cases} 2\pi e^{\omega} & \omega < 0 \\ 0 & \omega > 0 \end{cases}$$

$$(c) \quad x(t) = \int_{-\infty}^t \frac{\sin(\pi \tau)}{\pi \tau} d\tau$$

$$\frac{d}{dt} x(t) = \frac{\sin \pi t}{\pi t} \quad (*)$$

از طرفه دیگر: $\frac{\sin \pi t}{\pi t} \Leftrightarrow \text{rectangle}\left(\frac{\omega}{2\pi}\right) = \begin{cases} 1 & |\omega| < \pi \\ 0 & \omega = \pm \pi \end{cases}$

معمولاً مکزن دومانز: $\frac{dx(t)}{dt} \Leftrightarrow j\omega X(j\omega)$

$$\frac{(*) \text{ , } (**)}{|\omega| \neq 0} \Rightarrow j\omega X(j\omega) = \text{rectangle}\left(\frac{\omega}{2\pi}\right)$$

$$\Rightarrow X(j\omega) = \frac{1}{j\omega} \text{rectangle}\left(\frac{\omega}{2\pi}\right)$$

و برای حالت $\omega = 0$ و در نظر گرفتن آن :

$$\Rightarrow X(j\omega) = \pi \delta(\omega) + \frac{1}{j\omega} \text{rectangle}\left(\frac{\omega}{2\pi}\right)$$

سیگنال $y_c(t)$ حاصل کانولوشن دو سیگنال (C)

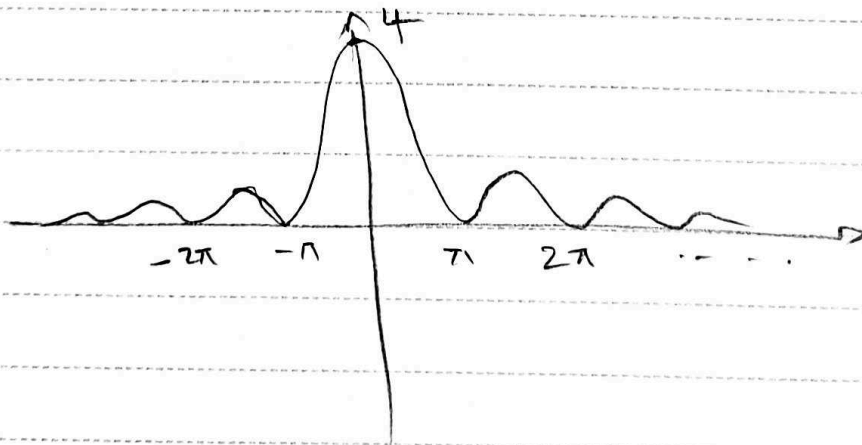
صاف با عرض $\frac{2}{\omega}$:

$$y_c(t) = x(t) * x(t)$$

و کانولوشن در حوزه زمان همان ضرب در حوزه

فرکانس است :

$$Y_c(j\omega) = X(j\omega) = \left(\frac{2 \sin \omega}{\omega} \right)^2$$



: 4 سوال

$$(a) \frac{d}{dt} y(t) + 3y(t) = x(t)$$

$$j\omega Y(j\omega) + 3Y(j\omega) = X(j\omega)$$

$$\Rightarrow (j\omega + 3)Y(j\omega) = X(j\omega)$$

$$H(j\omega) = \frac{Y(j\omega)}{X(j\omega)} = \frac{1}{j\omega + 3} \xleftrightarrow{F} e^{-3t} u(t)$$

$$\Rightarrow \boxed{h(t) = e^{-3t} u(t)}$$

$$(b) \frac{d^2}{dt^2} y(t) + 5 \frac{d}{dt} y(t) + 6y(t) = -\frac{d}{dt} x(t)$$

$$(j\omega)^2 Y(j\omega) + 5j\omega Y(j\omega) + 6Y(j\omega) = -j\omega X(j\omega)$$

$$\Rightarrow H(j\omega) = \frac{-j\omega}{(j\omega)^2 + 5j\omega + 6}$$

$$= \frac{A}{3+j\omega} + \frac{B}{2+j\omega}$$

$$\Rightarrow A = -3, B = 2$$

$$H(\omega) = \frac{-3}{3+j\omega} + \frac{2}{2+j\omega} \quad \leftarrow \boxed{h(t) = (-3e^{-3t} + 2e^{-2t})u(t)}$$

(a) $h(t) = 2e^{-2t}u(t) - 2te^{-2t}u(t)$ سؤال 5

$$2e^{-2t}u(t) \longleftrightarrow \frac{2}{2+j\omega}$$

$$2te^{-2t}u(t) \longleftrightarrow \frac{2}{(2+j\omega)^2}$$

$$\Rightarrow H(j\omega) = \frac{2}{2+j\omega} - \frac{2}{(2+j\omega)^2} = \frac{2+j2\omega}{(2+j\omega)^2}$$

$$= \frac{Y(j\omega)}{X(j\omega)}$$

$$\Rightarrow (4 + 4j\omega + (j\omega)^2) Y(j\omega) = (2+j2\omega) X(j\omega)$$

$$\Rightarrow \boxed{4y(t) + 4 \frac{d}{dt} y(t) + \frac{d^2}{dt^2} y(t) = 2x(t) + 2 \frac{d}{dt} x(t)}$$

$$(b) \quad H(j\omega) = \frac{2 + 3j\omega - 3(j\omega)^2}{1 + 2j\omega}$$

$$H(j\omega) = \frac{Y(j\omega)}{X(j\omega)} = \frac{2 + 3j\omega - 3(j\omega)^2}{1 + 2j\omega}$$

$$(1 + 2j\omega) Y(j\omega) = (2 + 3j\omega - 3(j\omega)^2) X(j\omega)$$

$$y(t) + 2 \frac{d}{dt} y(t) = 2x(t) + 3 \frac{d}{dt} x(t) - 3 \frac{d^2}{dt^2} x(t)$$